

VISHAL PATIL

Software Engineer | Node.js | Backend & AI Systems

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Nationality: Indian | Open to Relocation

PROFESSIONAL SUMMARY

Performance-driven Software Engineer with 4+ years of experience architecting scalable backend systems, AI-powered automation, and microservices for high-growth healthcare-tech platforms. Deep expertise in Node.js, MySQL, and AWS, spanning enterprise API gateway design, real-time conversational voice AI, multi-provider IVR systems, and payment orchestration — with a consistent record of delivering high-uptime, high-scale production systems (see Key Achievements below).

CORE TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Node.js (Express.js), React, Redux
AI & Voice	Google Vertex AI / Gemini, Gemini Vision, ElevenLabs TTS, Deepgram STT, XState, WebSockets, SSE, RAG Pipeline, Text Embeddings, Pinecone, Langfuse
Databases	MySQL, MongoDB, Redis, Sequelize ORM, Mongoose ORM
Cloud & DevOps	AWS (EC2, S3, Lambda), Firebase (FCM), PM2, New Relic, Grafana
Integrations	Knowlarity, Acephone, Xtremegen, Razorpay, Zoho CRM, Chat360, CMercury
Frontend	React, Redux, HTML, CSS, Tailwind CSS, Material-UI, Bootstrap
Tools & Security	GitHub, Winston, Morgan, JWT / RBAC, HMAC-SHA256 Webhooks, Rate Limiting

KEY ACHIEVEMENTS

36% → 72% Call Pickup Rate	85% Finance Workload Reduced
13% → 22% (+17%) IPD Conversion Rate	738+ endpoints / 19 microservices API Gateway Scale
~73% Lead Qualification Rate	99.9% System Uptime

PROFESSIONAL EXPERIENCE

Lead Software Engineer

Feb 2022 – Present

HexaHealth | India (Remote-friendly)

Enterprise API Gateway & Microservices Orchestration

- Designed a central API Gateway managing 738+ endpoints across 19 microservices (EHR, Scheduler, Payment, Notifications, CMS, VMS) with a middleware pipeline for JWT authentication and RBAC covering 20+ user roles.
- Built a security layer with intelligent rate limiting (120 req/min) and progressive IP blocking, maintaining 99.9% uptime while eliminating malicious traffic.
- Implemented full request observability using Winston and Morgan with auto-generated cURL commands, enabling 100% request traceability across all services.

Real-Time Conversational Voice AI & Campaign Automation

- Built a real-time speech-to-speech orchestration server using WebSockets, integrating Deepgram (STT), ElevenLabs (TTS), and Google Vertex AI/Gemini for bilingual (Hindi/English) patient consultations and structured lead qualification.
- Crafted a deterministic conversation state machine (XState) with 15+ nodes, handling complex branching, interruption, FAQ via RAG pipeline, and voice activity detection (VAD) for natural turn-taking.
- Engineered a production RAG pipeline for real-time FAQ resolution during voice calls — chunked and embedded healthcare knowledge-base content using Google text-embedding models, indexed vectors in Pinecone, and performed low-latency semantic search, reducing hallucinations and improving response accuracy.
- Established multi-provider IVR routing across Knowlarity, Acephone, and Xtremegen with dynamic provider selection based on performance averages, health checks, and intelligent fallback logic.
- Built an outbound dialer platform with slot-based calling, campaign queue strategies, priority-based retry, and DNP lifecycle management — lifted call pickup rate from 36% to 72% and achieved 73% lead qualification.

FinTech & Payment Orchestration

- Designed a production-grade Payment Microservice integrating Razorpay, supporting 15+ payment methods with end-to-end HMAC-SHA256-secured webhook verification.
- Built a dynamic pricing engine using complex MySQL stored procedures (300+ lines) supporting tiered (Gold/Silver/Bronze) and city-specific pricing overrides.

Healthcare EHR, Document AI & Vendor Management

- Built a scalable Electronic Health Record platform on AWS S3 + Lambda for serverless document processing (prescriptions, ID cards) with multi-resolution image generation.
- Integrated Google Gemini Vision for AI-driven medical bill parsing, extracting 20+ financial fields and accelerating insurance claim processing.
- Launched a multi-tenant Vendor Management System (VMS) with RBAC managing hospital-vendor agreements and rule-based revenue-sharing calculations across 50+ partners.
- Streamlined financial reconciliation via background jobs syncing payment status with Zoho CRM every 5 minutes, reducing manual finance workload by 85%.
- Expanded notification coverage with a unified multi-channel service (SMS, Email, WhatsApp, Push Notifications) supporting simultaneous delivery across all channels.

Full Stack Developer Intern

Jun 2021 – Feb 2022

Masai School | India

- Solved 800+ Data Structures and Algorithms problems; built production-quality full-stack website replicas (Discovery, Zoomcar, TravelTech) using React, Node.js, and MongoDB.
- Gained hands-on expertise in responsive design, Redux state management, and REST API integration.

NOTABLE PROJECTS

- **EC2 Log Management & Archival (Mar 2024):** Implemented scheduled log rotation and S3 archival, reducing EC2 disk usage from 98 GiB to 18 GiB and preventing server crashes through cron-based scheduling and retention policies.
- **User Management Service (Jan 2022):** Built a JWT-based authentication service with OTP and password login, bcrypt hashing, device logging, and secure session management.

EDUCATION

- **Bachelor of Technology** Aug 2018 – Jun 2021
Dr. Babasaheb Ambedkar Technological University, Lonere
First Class with Distinction | CGPA: 8.95 / 10
- **Diploma in Engineering** Aug 2015 – Jun 2018
Government Polytechnic, Miraj
First Class with Distinction, 85.18%

LANGUAGES

- English (Professional), Hindi (Native), Marathi (Native)